

Solutions

Maths

Q1. Let $y = (1 + x^2) \tan^{-1} x - x$ & $f(x) = \frac{dy}{dx}$, then $f(x) + \cot^{-1} x$ is equal to :

- (A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π

Ans. B

Sol. $\frac{dy}{dx} = 2x \tan^{-1} x + 1 - 1 = 2x \tan^{-1} x$

$$f(x) = \frac{dy}{dx} = 2x \tan^{-1} x$$

$$\text{then } \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

Q2. For the curve $32x^2y^2 = (x + y)^4$ the value of $\frac{d^2y}{dx^2}$ at (1, 1) is :

- (A) 0 (B) 1 (C) -1 (D) $\frac{1}{2}$

Ans. A

Sol. $32x^2y^2 = (x + y)^4$

$$\frac{dy}{dx} = \frac{y}{x} \text{ \& } \frac{d^2y}{dx^2} = 0 \text{ as homogeneous function.}$$

Q3. If $x = 2 + t^3$, $y = 3t^2$ and value of $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^n$ is constant, then value of n is :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. D

Sol. Given $x = 2 + t^3$, $y = 3t^2$

$$\frac{dx}{dt} = 3t^2, \frac{dy}{dt} = 6t$$

$$\frac{dy}{dx} = \frac{2}{t}$$

$$\frac{d^2y}{dx^2} = -\frac{2}{t^2} \left(\frac{dt}{dx}\right)$$

$$= -\frac{2}{t^2 \left(\frac{dx}{dt}\right)}$$

$$= -\frac{2}{3t^4}$$

$$\frac{\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^n} = \frac{-2}{3t^4 \left(\frac{2}{t}\right)^n}$$

$$\frac{d^2y}{dx^2}$$

is constant if $n = 4$

Q4. If $f(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$, then number of local extremas for $g(x)$, where $g(x) = f(|x|)$:

(A) 3 (B) 4 (C) 5 (D) None of these

Ans. C

Sol. $g(x) = \frac{(|x|-1)(|x|-2)}{(|x|-3)(|x|-4)}$

$g(x)$ is an even function so there is an extrema at $x = 0$.

Also number of extrema for $x > 0$ will be equal to number of extrema for $x < 0$

$$g(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$$

Number of extrema = 2

$\Rightarrow$ Total extrema = 5

Q5. The value of t for which $\frac{\int_0^{\pi/2} (\sin x + t \cos x) dx}{\sqrt{\int_0^{\pi/2} (\sin x + t \cos x)^2 dx}}$ is maximum lies in the interval :

(A) $\left(0, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{4}\right)$ (C) $\left(\frac{3}{4}, \frac{3}{2}\right)$ (D) $\left(\frac{3}{2}, \frac{5}{2}\right)$

Ans. C

Sol. Sol.

$$\int_0^{\pi/2} (\sin x + t \cos x) dx = 1 + t$$

$$\int_0^{\pi/2} (\sin x + t \cos x)^2 dx = \frac{\pi}{4}(1 + t^2) + t$$

$$y(t) = \frac{1+t}{\sqrt{\frac{\pi}{4}(1+t^2) + t}}$$

Let $\Rightarrow y^2(t)$

$$= \frac{(1+t)^2}{\frac{\pi}{4}(1+t^2) + t}$$

Note $y(t) > 0 \forall t \in (-1, \infty)$

And $y(t) \leq 0 \forall t \in (-\infty, 1]$

Hence, $y_{\max}$ will occur for some $t_0 > -1$

For $t > -1$

Maximum values of $y(t)$ and $y^2(t)$ will occur at same t_0

$$\begin{aligned} \frac{d}{dt}(y^2(t)) &= \frac{2(t+1)}{\frac{\pi}{4}(1+t^2) + t} - \frac{(t+1)^2(\frac{\pi}{2}t+1)}{(\frac{\pi}{4}(1+t^2) + t)^2} \\ &= \frac{(1+t)[\frac{\pi}{2}(1+t^2) + 2t - (t+1)(\frac{\pi}{2}t+1)]}{(\frac{\pi}{4}(1+t^2) + t)^2} \end{aligned}$$

$$= \frac{(1+t)(1-t)(\frac{\pi}{2}-1)}{(\frac{\pi}{4}(1+t^2) + t)^2}$$

$\Rightarrow$

$y^2(t)$ has maximum at $t = 1$ and minimum at $t = -1$

$\Rightarrow$

$$t_0 = 1$$

- Q6. $f(x) = |x - 1| + |x - 2|$, $I = \int_0^3 f(x)dx$, M = the minimum value of f , $N = f'(x)$ for $x < -4$ and C = the value of $f''(4)$. Then the value of $\frac{M^2 - N^2 + IC}{2}$ is :

(A) $-\frac{3}{2}$ (B) $-\frac{5}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. A

Sol.
$$f(x) = \begin{cases} 3 - 2x & x < 1 \\ 1 & 1 \leq x < 2 \\ 2x - 3 & x \geq 2 \end{cases}$$

Clearly, $I = 5$ {Using the piece wise definition of f }

$$M = 1$$

$$N = -2$$

$$C = 0$$

- Q7. If $f(x)$ is a function such that $f(x) + f''(x) = 0$ and $g(x) = (f(x))^2 + (f'(x))^2$ and $g(3) = 8$, then $g(8) =$:
- (A) 0 (B) 3 (C) 5 (D) 8

Ans. D

Sol. $g(x) = (f(x))^2 + (f'(x))^2 \Rightarrow g'(x) = 2f(x)f'(x) + 2f'(x)f''(x)$
or $g'(x) = 2f(x)f'(x) - 2f'(x)f(x) = 0 \Rightarrow g(x) = c \Rightarrow g(8) = 8$

- Q8. Let f and g be differentiable functions on $\mathbb{R}$ (the set of all real numbers) such that $g(1) = 2 = g'(1)$ and $f'(0) = 4$. If $h(x) = f(2xg(x) + \cos \pi x - 3)$ then $h'(1)$ is equal to :
- (A) 28 (B) 24 (C) 32 (D) 18

Ans. C

Sol. $h(x) = f(2xg(x) + \cos \pi x - 3)$
 $h'(x) = f'(2xg(x) + \cos \pi x - 3)[2g(x) + 2xg'(x) - \pi \sin \pi x]$
 $h'(1) = f'(2g(1) - 4)[2g(1) + 2g'(1)] = 32$

- Q9. If $f(x) = \frac{(x+1)^7 \sqrt{1+x^2}}{(x^2-x+1)^6}$, then the value of $f'(0)$ is equal to :

(A) 10 (B) 11 (C) 13 (D) 15

Ans. C

Sol. $f(x) = \frac{(x+1)^7 \sqrt{1+x^2}}{(x^2-x+1)^6}$ ($f(0) = 1$)
 $\ln f(x) = 7 \ln(1+x) + \frac{1}{2} \ln(1+x^2) - 6 \ln(x^2-x+1)$
$$\frac{f'(x)}{f(x)} = \frac{7}{1+x} + \frac{x}{1+x^2} - \frac{6(2x-1)}{x^2-x+1}$$

$$f'(0) = 13$$

Q10. Let f, g be differentiable functions on the real line $\mathbb{R}$ with $f(0) > g(0)$. Assume that the set $M = \{t \in \mathbb{R} \mid f(t) = g(t)\}$ is non-empty and that $f'(t) \geq g'(t)$ for all $t \in M$. Then which of the following is necessarily true ?

- (A) if $t \in M$, then $t < 0$ (B) for any $t \in M$, $f'(t) > g'(t)$ (C) For any $t \notin M$, $f(t) > g(t)$
 (D) None of these

Ans. D

Sol. Let $f(t) = (t+1)(t-1)^2$, $g(t) = 0$. Then $f(0) = 1 > g(0)$.

$$M = \{t \in \mathbb{R} : f(t) = g(t)\} \\ = \{t \in \mathbb{R} : (t+1)(t-1)^2 = 0\}$$

The set $= \{-1, 1\}$ is not empty. $\setminus (f'(t), g'(t) = 0$.

$$= (t-1)^2 + 2(t+1)(t-1) \\ = (t-1)(2t+1)$$

$$f'(-1) = 2 > 0 = g'(-1), f'(1) = 0 = g'(1)$$

f and g satisfy the conditions of the problem. However, neither of (A), (B), (C) is true.

Q11. If the normal to the curve $y = f(x)$ at $x = 0$ be given by $3x - y + 3 = 0$, then the value of $\lim_{x \rightarrow 1} (x-1)^2 \{f((x-1)^2) - 5f(4(x-1)^2) + 4f(7(x-1)^2)\}^{-1}$ is:

- (A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 3 (D) -3

Ans. A

Sol. Slope of normal = 3

$$\Rightarrow \text{slope of tangent } f'(0) = -\frac{1}{3}$$

$$L = \lim_{x \rightarrow 1} \frac{(x-1)^2}{f((x-1)^2) - 5f(4(x-1)^2) + 4f(7(x-1)^2)} \left(\frac{0}{0} \text{ form} \right) \\ = \lim_{x \rightarrow 1} \frac{(x-1)^2}{\frac{2(x-1)f'((x-1)^2) - 40f'(4(x-1)^2)(x-1) + 56f'(7(x-1)^2)(x-1)}{2}} \\ = \lim_{x \rightarrow 1} \frac{1}{2f'((x-1)^2) - 40f'(4(x-1)^2) + 56f'(7(x-1)^2)} \\ = \frac{1}{f'(0) - 20f'(0) + 28f'(0)} \\ = \frac{1}{-\frac{1}{3} - 20(-\frac{1}{3}) + 28(-\frac{1}{3})} \\ = \frac{1}{-\frac{1}{3} + \frac{20}{3} - \frac{28}{3}} \\ = -\frac{1}{3}$$

Q12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any twice differentiable function such that its second derivative is continuous and $\frac{df(x)}{dx} \neq 0$ for all $x \neq 0$. If then $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \pi$:

- (A) for all $x \neq 0$, $f(x) > f(0)$ (B) for all $x \neq 0$, $f(x) < f(0)$ (C) for all x , $\frac{d^2f(x)}{dx^2} > 0$

(D) for all x , $\frac{d^2 f(x)}{dx^2} < 0$

Ans. A

Sol. Observe that $f(0) = 0$ (otherwise $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$ does not exist in $\mathbb{R}$). Since $\pi > 1$, there exists $\epsilon > 0$ such that

$$x \in (-\epsilon, \epsilon) \setminus \{0\} \Rightarrow f(x) > x^2$$

On the set $(0, \infty)$ we have $f'(x) \neq 0$. So, the derivative $f'(x)$ preserves the sign on $(0, \infty)$. If

$x \in (0, \epsilon)$, then relations $f(0) = 0$, $f(x) > 0$ imply that f' is positive somewhere on $(0, x)$. But then it is positive everywhere on $(0, \infty)$ and $f(x) > f(0)$ for $x > 0$.

On the set $(-\infty, 0)$ we have $f'(x) \neq 0$. So, the derivative $f'(x)$ preserves the sign on $(-\infty, 0)$. If $x \in (-\epsilon, 0)$, then relations $f(x) > 0$, $f(0) = 0$ imply that f' is negative somewhere on $(x, 0)$. But then it is negative everywhere on $(-\infty, 0)$ and $f(x) > f(0)$ for $x < 0$.

Q13. A factory D is to be connected by a road with a straight railway line on which a town A is situated. The distance DB of the factory to the railway line is $5\sqrt{3}$ km. Length AB of the railway line is 20 km. Freight charges on the road are twice the charges on the railway. The point P ($AP < AB$) on the railway line should the road DP be connected so as to ensure minimum freight charges from the factory to the town is :

- (A) BP = 5km (B) AP = 5km (C) BP = 7.5 km (D) None of these

Ans. A

Sol. Let the charges for railway line be k Rs/km.

Now, total freight charges, $T = kx + 2k\sqrt{(20-x)^2 + 75}$

$$\text{Let } \frac{dT}{dx} = 0$$

$$\text{Or } k + 2k \frac{2(x-20)}{2\sqrt{(x-20)^2 + 75}} = 0$$

$$\text{Or } 4(x-20)^2 = 75 + (x-20)^2$$

$$\text{Or } (x-20)^2 = 25$$

$$\text{Or } x = 25, 15 \text{ or } x = 15 \text{ (as } AP < AB)$$

$$\text{Or } PB = AB - AP = 20 - 15 = 5\text{km}$$

Q14. Let $f(x) = \sin^{-1} \left(\frac{2g(x)}{1+g(x)^2} \right)$, then which are correct?

- (i) $f(x)$ is decreasing if $g(x)$ is increasing and $|g(x)| > 1$
 (ii) $f(x)$ is an increasing function if $g(x)$ is increasing and $|g(x)| \leq 1$
 (iii) $f(x)$ is decreasing function if $g(x)$ is decreasing and $|g(x)| > 1$
 (A) (i) and (iii) (B) (i) and (ii) (C) (i), (ii) and (iii) (D) (iii)

Ans. B

Sol. $f(x) = 2 \tan^{-1}(g(x))$ $|g(x)| \leq 1$

$$= \pi - 2 \tan^{-1} g(x) \quad g(x) > 1$$

$$= -\pi - 2 \tan^{-1} g(x) \quad g(x) < -1$$

$$f'(x) = \frac{2g'(x)}{1 + (g(x))^2} |g(x)| < 1$$

$$= -\frac{2g'(x)}{1 + (g(x))^2} |g(x)| > 1$$

Q15. For the curve $y = x^2 - x + 1$ & $x = y^2 - y + 1$ which of the following is/are true :

- (A) $y = x$ is their common tangent (B) $y + x = 2$ is their common normal
(C) both the curves touch each other (D) both the curves never touch each other

Ans. A, B, C

Sol. $y = x^2 - x + 1$ _____ (1)

$x = y^2 - y + 1$ _____ (2)

from eq. (1) and (2)

$$x = y(y - 1) + 1$$

$$x = (x^2 - x + 1)(x^2 - x) + 1$$

$$x = (x^2 - x)^2 + x^2 - x + 1$$

$$x^2(x - 1)^2 + (x - 1)^2 = 0$$

$$(x - 1)^2(x^2 + 1) = 0$$

Curve touch each other at $x = 1$

at $x = 1, y = 1$

Tangent at (1, 1) is $y = x$

Normal at (1, 1) is $x + y = 2$

Q16. If $f(x) = \ln((\cos x)^{\cos x} + 1)$, $x \in \left(0, \frac{\pi}{2}\right)$ then which of the following is true ?

(A) $f(x)_{\min} = \ln\left(\left(\frac{1}{e}\right)^{\frac{1}{e}} + 1\right)$ (B) $f(x)_{\max} = \infty$ (C) $f(x)_{\min} = \log 2$

(D) $y \in \left[\ln\left(\left(\frac{1}{e}\right)^{\frac{1}{e}} + 1\right), \ln 2\right]$

Ans. A, D

Sol. Let $t = \cos x$. Since $x \in \left(0, \frac{\pi}{2}\right)$, we have $t \in (0, 1)$.

Then the function becomes $f(x) = \ln(t^t + 1)$.

Because $\ln$ is an increasing function, the extrema of $f(x)$ occur where t^t has extrema.

$$\text{Let } \phi(t) = t^t = e^{t \ln t}.$$

Differentiating, $\phi'(t) = t^t(\ln t + 1)$.

Thus $\phi'(t) = 0$ gives $\ln t = -1$, so $t = \frac{1}{e}$.

For $0 < t < \frac{1}{e}$, $\phi'(t) < 0$, so $\phi(t)$ is decreasing.

For $\frac{1}{e} < t < 1$, $\phi'(t) > 0$, so $\phi(t)$ is increasing.

Hence $\phi(t)$ has a minimum at $t = \frac{1}{e}$.

The minimum value is $\phi\left(\frac{1}{e}\right) = \left(\frac{1}{e}\right)^{\frac{1}{e}}$.

Therefore $f(x)_{\min} = \ln\left(\left(\frac{1}{e}\right)^{\frac{1}{e}} + 1\right)$, so option A is true.

Also, on $(0, 1)$, $t^t < 1$, so $f(x) < \ln 2$.

As $t \rightarrow 1^-$, $t^t \rightarrow 1$, hence $f(x) \rightarrow \ln 2$, but this value is not attained because $x < \frac{\pi}{2}$.

So $f(x)$ does not have a finite maximum; option B is false.

Option C is false because the minimum is not $\log 2$.

For all admissible x , $y = f(x)$ satisfies $\ln\left(\left(\frac{1}{e}\right)^{\frac{1}{e}} + 1\right) \leq y < \ln 2$.

Thus y also belongs to the closed interval $\left[\ln\left(\left(\frac{1}{e}\right)^{\frac{1}{e}} + 1\right), \ln 2\right]$, so option D is correct in the given key, with the note that the upper endpoint is only approached, not attained.

Therefore the correct options are **A** and **D**.

Q17. Which of the following function(s) $y = f(x)$ cannot exist ?

- (A) $\left(\frac{d^2y}{dx^2}\right) > 0; \forall x \in R, \left(\frac{dy}{dx}\right)_{x=0} = \left(\frac{dy}{dx}\right)_{x=1} = 1$
- (B) $\left(\frac{d^2y}{dx^2}\right) > 0; \forall x \in R, \left(\frac{dy}{dx}\right)_{x=0} = 15; \left(\frac{dy}{dx}\right)_{x=1} = 11$
- (C) $\left(\frac{d^2y}{dx^2}\right) \geq 0; \forall x \in R; f'(0) = \frac{1}{2017}$ and $\lim_{x \rightarrow \infty} f(x) = 2017$
- (D) $\frac{d^2y}{dx^2} > 0, \forall x \in R; \left(\frac{dy}{dx}\right)_{x=0} = 1$ and $f(x) < 1$ for all $x < 0$

Ans. A, B, C

Sol. (A) $\frac{d^2y}{dx^2} > 0$
 $\implies f'(x)$ is increasing thus $f'(0) \neq f'(1)$

(B) $f'(1) > f'(0)$

Thus B is not possible (as $f'(x)$ is increasing function)

(C) $\frac{d^2y}{dx^2} \geq 0$ mean graph must be concave up

& $f'(0) = \frac{1}{2017}$

for $\lim_{x \rightarrow \infty} f(x) = 2017$ graph has to change concavity, which is not possible.

(D) There exists such function satisfying given condition e.g. $f(x) = e^x$

Q18. The function $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 2\lambda x + 5$ has a maximum and a minimum for :

- (A) $\lambda \in (-4, \infty)$ (B) $\lambda \in (-\infty, 0)$ (C) $\lambda \in (-3, 3)$ (D) $\lambda \in (1, \infty)$

Ans. A, B, C, D

Sol. $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 2\lambda x + 5$
 $f'(x) = 6x^2 - 6(\lambda + 2)x + 2\lambda = 0$ has two real roots, then
 $D > 0 \Rightarrow 3\lambda^2 + 8\lambda + 12 > 0$
 $\Rightarrow \lambda \in R$

Q19. Let m and n be positive integers and $x, y > 0$ and $x + y = k$, where k is constant. Let $f(x, y) = x^m y^n$, then :

- (A) $f(x, y)$ is maximum when $x = \frac{mk}{m+n}$ (B) $f(x, y)$ is maximum where $x = y$
(C) maximum value of $f(x, y)$ is $\frac{m^m n^n k^{m+n}}{(m+n)^{m+n}}$ (D) maximum value of $f(x, y)$ is $\frac{k^{m+n} m^m n^n}{(m+n)^{m+n}}$

Ans. A, D

Sol. $f(x, y) = x^m (k - x)^n$
 $f'(x, y) = mx^{m-1}(k - x)^n - x^m n \cdot (k - x)^{n-1} = 0$
 $\Rightarrow x = \frac{mk}{m+n}$
Maximum value = $\frac{k^{m+n} m^m n^n}{(m+n)^{m+n}}$

Q20. If $f(x) = 2^x + x^3 + 1$ and $g(x) = f^{-1}(x)$, then : (where dash (') denotes derivative)

- (A) $g'(2) = 1$ (B) $g'(2) = \log_2 e$ (C) $g''(2) = (\ln 2)^2$ (D) $g''(2) = -\log_2 e$

Ans. B, D

Sol. $f(x) = 2^x + x^3 + 1$
 $\Rightarrow f'(x) = 2^x \ln 2 + 3x^2$
 $\Rightarrow f''(x) = 2^x (\ln 2)^2 + 6x$
 $f(0) = 2, g(g(x)) = x$
 $\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$
 $\Rightarrow g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$
 $g''(2) = \frac{1}{f'(0)} = \frac{1}{\ln 2} = \log_2 e$
 $g''(2) = -\log_2 e$

Q21. If $f(x) = \frac{1}{x^2 - 1}$, then identify the **CORRECT** option(s) (where $f^n(x)$ represent $\frac{d^n f(x)}{dx^n}$:

- (A) $f^{2013}(x)$ is an odd function (B) $f^{2014}(x)$ is an even function
(C) $f^n(x)$ is always an odd function (D) $f^n(x) = \frac{(-1)^n n!}{2} \left[\frac{(x-1)^{n-1} - (x+1)^{n-1}}{(x^2-1)^{n-1}} \right]$

Ans. A, B, D

Sol. $f(x) = \frac{1}{x^2 - 1} \Rightarrow f(x)$ is an even function"its derivative is an odd function

$f^{2013}(x)$ is an odd function

$f^{2014}(x)$ is an even function

$$f(x) = \frac{1}{2} \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$$

$$f^n(x) = \frac{(-1)^n \cdot n!}{2} \left(\frac{(x+1)^{n+1} - (x-1)^{n+1}}{(x^2-1)^{n+1}} \right)$$

Q22. Let $y = f(x)$ be a differentiable function such that $f'(x) > 0 \forall x \in R$ then which of the following is/are correct ?

(A) $f(3-4x)$ is an increasing function (B) $f(3-4x)$ is a decreasing function

(C) $f(x^2-x)$ is increasing $\forall x > \frac{1}{2}$ (D) $y = (f(x))^3$ is an increasing function

Ans. B, C, D

Sol. $\frac{d}{dx} f(3-4x) = -4f'(3-4x) < 0$; $\frac{d}{dx} (f(x))^3 = 3f'(x)(f(x))^2 > 0$
 $\frac{d}{dx} f(x^2-x) = (2x-1)f'(x^2-x) > 0$ if $x > \frac{1}{2}$

Q23. Let $f(x)$ be a differentiable function with $f(1)f(-1) \neq 0$. Define a function $g(x) = \frac{x^2-1}{f(x)}$. If $g(x)$ does not follow Rolle's theorem in $[-1, 1]$, then which of the following options is/are false ?

(A) $f(x) = 0$ cannot have any root in $[-1, 1]$ (B) $f(x) = 0$ has at least one root in $[-1, 1]$

(C) $f'(x)$ is zero at atleast one point in $[-1, 1]$ (D) $f(x)$ cannot satisfy Rolle's theorem in $[-1, 1]$

Ans. A, C, D

Sol. $g(x)$ is not necessarily be defined in the complete interval $[-1, 1]$ and $g(x)$ will satisfy Rolle's theorem unless $f(x) = 0$ for some $x \in (-1, 1)$.

So far as $f'(x)$ is concerned, without further information nothing definite can be said.

Q24. Let a function ' f ' is continuous and differentiable with $f(0) = 1$ and $f(1) = 2$. Then, for $\forall n \in N, n \geq 2$, there exists atleast one $c \in [0, 1]$ such that :

(A) $(f'(c)[f(c)]^{n-1} > \sqrt{2^{n-1}}$ (B) $(f'(c)[f(c)]^{n-1} < \sqrt{2^{n-1}}$ (C) $f'(c)[f(c)]^{n-1} > 1$

(D) $f'(c)[f(c)]^{n-1} < 1$

Ans. A, C

Sol. Let $g(x) = [f(x)]^n$

As $g(x)$ is continuous and differentiable in $[0, 1]$

$\therefore$ By LMVT, there exists a $c \in [0, 1]$, such that $g'(c) = \frac{f(1) - f(0)}{1 - 0}$

$$\Rightarrow n[f(x)]^{n-1} f'(x) \Big|_{x=c} = \frac{2^n - 1}{1}$$

$$\Rightarrow n f'(c) \cdot [f(c)]^{n-1} = 1 + 2 + 2^2 + \dots + 2^{n-1}$$

$$\Rightarrow f'(c)[f(c)]^{n-1} = \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n}$$

As $AM \geq GM$,

$$\begin{aligned} \therefore \frac{1+2+2^2+\dots+2^{n-1}}{n} &\geq \sqrt[n]{2^{1+2+\dots+n-1}} \\ \Rightarrow \frac{1+2+2^2+\dots+2^{n-1}}{n} &> 2^{\frac{(n-1)(n)}{2} \cdot \frac{1}{n}} \\ \therefore f'(c)[f(c)]^{n-1} &> 2^{\frac{n-1}{2}} \end{aligned}$$

Q25. A stationary point of a function f is a real number r such that $f'(r) = 0$. Consider a polynomial $p(x)$.

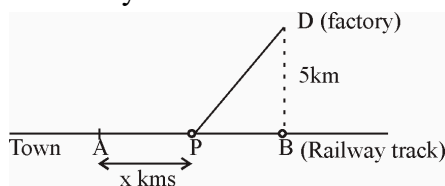
- (A) If $p(x)$ is of degree 2022, then $p(x)$ must have at least one stationary point.
 (B) If the number of distinct real roots of $p(x)$ is 2021, then $p(x)$ must have at least 2020 stationary points.
 (C) If the number of distinct real roots of $p(x)$ is 2021, then $p(x)$ can have at most 2020 stationary points.
 (D) If r is a stationary point of $p(x)$ AND $p''(r) = 0$, then the point $(r, p(r))$ is neither a local maximum nor a local minimum point on the graph of $p(x)$.

Ans. A, B

- Sol. (a) $p'(x)$ is a polynomial of degree 2021, which is odd, so it has a root by intermediate value theorem by looking at behaviour as $x \rightarrow \pm\infty$.
 (b) The graph of $p(x)$ has to turn between any two consecutive zeros, giving a stationary point, in fact a local max/min
 (c) The graph of $p(x)$ can turn more than once between zeros, or turn outside extreme zeros or have stationary points that are not maxima or minima.
 (d) is false, e.g., $p(x) = x^4$.

Paragraph

A factory is connected with a straight railway track by a highway. Town A lies on the railway track. The distance DB from the factory to railway track is 5 km. The segment AB of railway track is equal to ' ℓ ' kms. Freight charges on highway are 2 times higher than on railway. A highway DP is connected to railway track such that AP is x kms (see the figure).



Q26. If $\ell = \frac{185}{\sqrt{3}}$ kms, then the value of ' x ' for which freight charge is minimum :

- (A) $\frac{60}{\sqrt{3}}$ kms (B) $60\sqrt{3}$ kms (C) $120\sqrt{3}$ kms (D) $75\sqrt{3}$ kms

Ans. B

Sol. $f(x) = x + 2\sqrt{25 + (\ell - x)^2} \dots(i)$
 $0 \leq x \leq \ell$
 $f'(x) = 1 - \frac{2(\ell - x)}{\sqrt{25 + (\ell - x)^2}} \dots(ii)$
 $= \frac{\sqrt{25 + (\ell - x)^2} - 2(\ell - x)}{\sqrt{25 + (\ell - x)^2}}$

$$\begin{aligned}
25 + (\ell - x)^2 &= 4(\ell - x)^2 \\
\sqrt{3}(\ell - x) &= \pm 5 \\
x &= \ell \pm \frac{5}{\sqrt{3}} \\
x &= \ell - \frac{5}{\sqrt{3}} \\
\text{If } \ell &= \frac{185}{\sqrt{3}} \text{ then } x = \frac{180}{\sqrt{3}} = 60\sqrt{3} \text{ kms}
\end{aligned}$$

Q27. If $\ell = 10$ kms, then the value of x for which freight charges are maximum :

- (A) 10 kms (B) $10 - \frac{5}{\sqrt{3}}$ kms (C) $\frac{10}{\sqrt{3}}$ kms (D) 0 km

Ans. D

Sol. $f(x) = x + 2\sqrt{25 + (\ell - x)^2} \dots(i)$
 $0 \leq x \leq \ell$
 $\ell = 10$ kms, then maximum value of $f(x)$ lies at $x = 0$ or $x = 10$
 $f(x) = x + 2\sqrt{25 + (10 - x)^2}$
 $f(0) = 10\sqrt{5}$
 $f(10) = 20$
 $\Rightarrow$ maximum freight = $10\sqrt{5}$ at $x = 0$.

Paragraph

A function $f : R \rightarrow R$ satisfies the following properties :

$$f'(x) > 0 \forall x \in (-\infty, -2) \cup (3, \infty), f'(x) \leq 0 \forall x \in (-2, 3), \lim_{x \rightarrow 2} f(x) = 0$$

f is continuous $\forall x \in R - P\{3\}$, f is differentiable $\forall x \in R - \{-2, 3\}$, $\lim_{x \rightarrow 3} f(x) \rightarrow -\infty$

$f''(x) > 0 \forall x \in (-\infty, 0) - \{-2\}$, $f''(x) < 0 \forall x \in (0, \infty) - \{3\}$, $\lim_{x \rightarrow -\infty} f(x) = 3$, and $f(0) = 0$. Now,

answer the following questions :

Q28. Let λ be the maximum number of solutions of $f(x) = |x|$. If $f'(0) > -3$ and $f(-2) > 6$, then number of solutions of $f(x) + 3x = 0$ is equal to :

- (A) $\lambda + 1$ (B) $\lambda - 1$ (C) $2\lambda - 1$ (D) $3\lambda - 4$

Ans. A

Sol. -

Q29. The function $y = f(-|x|)$ is :

- (A) differentiable $\forall x \in R$, iff $f'(0) = 0$
 (B) continuous but NOT differentiable at two points, iff $f'(0) = 0$
 (C) continuous but NOT differentiable at one point, iff $f'(0) = 0$
 (D) dis-continuous at two points if $f'(0) = 0$

Ans. B

Sol. -

Q30. Let $f(x)$ be a polynomial of degree 2016 satisfying $f(1) = \frac{1}{2}, f(2) = 2, f(3) = -5, f(-4) = 3$. If $f(x)$ be an even function and minimum number of points of inflection of $f(x)$ is n , then the value of $\frac{1}{n}$ is _____.

Ans. 0.25

Sol. $\therefore f'(\alpha) = f'(\beta) = f'(-\alpha) = f'(-\beta) = f'(0) = 0$
 $\therefore$ using Rolle's theorem minimum number of points of inflection are four. 

Q31. A cuboid, length of whose one edge is $\sqrt{2}$ units, is inscribed in a sphere of radius $\sqrt{2}$ units. If maximum possible volume of cuboid is V cubic units, then $\frac{V}{2\sqrt{2}}$ is _____.

Ans. 1.50

Sol. Let the other edges be 'a' & 'b'

$$\left(\frac{\sqrt{a^2 + b^2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 = 2$$
 (pythagoruous theorem)

$$\frac{a^2 + b^2}{4} + \frac{2}{4} = 2 \Rightarrow a^2 + b^2 = 6$$
 Now, volume cuboid $= ab\sqrt{2}$

$$\therefore \frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2}$$

$$\frac{a^2 + b^2}{2} \geq ab \Rightarrow ab \leq 3$$

$$\therefore V_{\max} = 3\sqrt{2}$$

$$\frac{V}{2\sqrt{2}} = \frac{3\sqrt{2}}{2\sqrt{2}} = 1.5$$
 

Q32. The greatest value of the function $f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1}$ is:

Ans. 16

Sol. We have
$$f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1}$$

$$= \frac{x^4 + 4x^3 + 6x^2 + 4x + 1}{x^4 - x^3 + x^2 - x + 1}$$

$$= \frac{x^2(x^2 + \frac{1}{x^2}) + 4x(x + \frac{1}{x}) + 6}{x^2(x^2 + \frac{1}{x^2}) - x(x + \frac{1}{x}) + 1}$$

$$= \frac{(x^2 + \frac{1}{x^2}) + 4(x + \frac{1}{x}) + 6}{(x^2 + \frac{1}{x^2}) - (x + \frac{1}{x}) + 1}$$
 Let $g(x) = \left(x + \frac{1}{x}\right), g'(x) = \left(1 - \frac{1}{x^2}\right)$
 $g'(x) = 0$ gives $\left(1 - \frac{1}{x^2}\right) = 0$
 $x = \pm 1$; Hence, the greatest value of $f(x)$ is $f(1) = \frac{2+8+6}{2-2+1} = 16$

Q33. The number of distinct real roots of equation $2x^4 - 8x^3 + 8x^2 - 1 = 0$ is n , then the value of $\frac{1}{n}$ is _____.

Ans. 0.25

Sol.  Sol.

$$f(x) = 2x^4 - 8x^3 + 8x^2 - 1$$

$$f'(x) = 8x^3 - 24x^2 + 16x$$

$$= 8x(x^2 - 3x + 2)$$

$$= 8x(x-1)(x-2)$$

$$f(0) = -1, f(1) = 1, f(2) = -1$$

So, $f(x)$ has 4 distinct roots.

Q34. Let $f(x)$ be a non-constant thrice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(6-x)$ and $f'(0) = 0 = f'(2) = f'(5)$. If n is the minimum number of roots of $(f''(x))^2 + f'(x)f'''(x) = 0$ in the interval $[0, 6]$, then the value of n is _____.

Ans. 12

Sol. On differentiating (1) w.r.t. x , we get

$$f'(x) = -f'(6-x)$$

Putting $x = 0, 2, 3, 5$ in (2), we get

$$f'(0) = -f'(6) = 0$$

$$\text{Similarly, } f'(2) = f'(4) = 0$$

$$f'(3) = 0$$

△ Possibly misplaced — please review:

$$f(5) = f(1) = 0$$

$$\therefore f'(0) = 0 = f'(2) = f'(3) = f'(5) = f'(1) = f'(4) = f'(6)$$

Therefore, $f'(x) = 0$ has minimum seven roots in $[0, 6]$

Now, consider a function $y = f''(x)$.

As $f''(x)$ satisfies Rolle's theorem in intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, $[3, 4]$, $[4, 5]$ and $[5, 6]$, respectively,

by Rolle's theorem, the equation $f'''(x) = 0$ has minimum six roots.

Now, $g(x) = (f''(x))^2 + f'(x)f'''(x) = 0 = h'(x)$, where

$$h(x) = f'(x)f''(x).$$

Clearly, $h(x) = 0$ has minimum 13 roots in $[0, 6]$.

Q35. If $f(g(x)) = 4 - x + 5x^{1/3} - x^{2/3}$ and $g(x) = 2 - x^{1/3}$ then local maxima value of $f(x)$ is equal to

Ans. 7

Sol. Let $x^{1/3} = 2 - g(x)$

$$\Rightarrow f(x) = 4 - (2-x)^3 + 5(2-x) - (2-x)^2$$

$$f(x) = x^3 - 7x^2 + 11x + 2$$

$$f'(x) = 0 \Rightarrow x = 1, \frac{11}{3}$$

$$f''(1) < 0$$

at $x = 1$, $f(x)$ is maximum

$$f(x)_{\max.} = 7$$

Q36. If $f : [2, \infty) \rightarrow R$; $f(x)$ is a function which is differentiable as well as increasing & $f(2) = 2$; then the number of solution of equation $f(f(x)) = \frac{4}{(x^2 - 4x + 6)}$ is _____.

Ans. 1

Sol.  Sol.

$$f(f(x))$$

is also an increasing function

$$\text{Let } y = \frac{4}{(x^2 - 4x + 6)}$$

$$\frac{dy}{dx} = -\frac{8(x-2)}{(x^2 - 4x + 6)^2}$$

is decreasing function.

Only one solution is possible

Q37. Raghav and his father are standing together on a circular track of radius 100 meters. When father gives a signal Raghav starts to run around the track at a speed of 10m/s. If the displacement of Raghav from his father is increasing at the rate of $\ell\sqrt{2}$ m/s when Raghav has run $\frac{1}{4}$ of the way around the track then ℓ is _____.

Ans. 5

Sol. $x^2 = 100^2 + 100^2 - 2(100)^2 \cos \theta$

$$= 2 \cdot 100^2 [1 - \cos \theta] \quad \text{$$

$$x^2 = 100^2 + 100^2 - 2(100)^2 \cos \theta = 2 \cdot 100^2 [1 - \cos \theta]$$

$$2x \frac{dx}{dt} = 2 \cdot 10^4 \sin \theta \frac{d\theta}{dt}$$

$$s = 100\theta$$

$$\Rightarrow \frac{ds}{dt}$$

Also $= 100 \frac{d\theta}{dt}$

$$\Rightarrow \frac{d\theta}{dt}$$

$$= \frac{1}{10}$$

(2) and $\theta = \frac{\pi}{2}$ in (1)

$$\Rightarrow \frac{dx}{dt} = \frac{2 \cdot 10^4 \cdot 1}{2 \cdot 100 \sqrt{2} \cdot 10} = 5\sqrt{2}$$

Q38. Let the maximum value of the expression $y = \frac{x^4 - x^2}{x^6 + 2x^3 - 1}$ for $x > 1$ is $\frac{p}{q}$ where p and q are relatively prime positive integers. The value of $(p + q)$ is k. For a unique value of μ & λ , the system of equations given by $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 5y + \lambda z = \mu$ has infinitely many solutions, then $\frac{\mu - \lambda}{4} - k$ is equal to _____.

Ans. 0

Sol. $y = \frac{x - \frac{1}{x}}{x^3 - \frac{1}{x^3} + 2}$

Let $t = x - \frac{1}{x} > 0$ for $x > 1$

$$\begin{aligned}
y &= \frac{t}{t(t^2 + 3) + 2} \\
x^3 - \frac{1}{x^3} &= t(t^2 + 3) \\
&= \frac{t}{t^3 + 3t + 2} = \frac{1}{t^2 + \frac{2}{t} + 3} \\
t^2 + \frac{2}{t} &= t^2 + \frac{1}{t} + \frac{1}{t} \geq 3 \\
\therefore t^2 + \frac{2}{t} + 3 &\geq 6 \text{ (AM} \geq \text{GM)} \\
y_{\max} &= \frac{1}{(t^2 + \frac{2}{t} + 3)_{\min}} = \frac{1}{6} \\
p &= 1, q = 6 \\
D &= \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 5 & \lambda \end{vmatrix} = \lambda - 8 = 0 \Rightarrow \lambda = 8 \\
D_3 &= \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 14 \\ 2 & 5 & \mu \end{vmatrix} = \mu - 36 = 0 \Rightarrow \mu = 36
\end{aligned}$$

- Q39. The value of $7 + \frac{2}{\pi} \sin^{-1} [\cos \{ \cos^{-1}(\cos x) + \sin^{-1}(\sin x) \}]$ where $\frac{\pi}{2} < x < \pi$ is k.
A differentiable function f satisfying a relation $f(x+y) = f(x) + f(y) + 2xy(x+y) - \frac{1}{3} \forall x, y \in R$
and $\lim_{h \rightarrow 0} \frac{3f(h) - 1}{kh} = \frac{2}{3}$. Find the value of definite integral $\int_{-3}^3 f(x) dx$ _____.

Ans. 2

Sol. Sol.

$$\begin{aligned}
\frac{f^c}{d}(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{f(h) - \frac{1}{3}}{h} \\
&= 2x(x+h) \\
\Rightarrow \frac{f^c}{d}(x) &= \\
&= \frac{4}{3} + 2x^2 \\
\Rightarrow f(x) &= \\
&= \frac{4}{3}x + \frac{2}{3}x^3 + \frac{1}{3}
\end{aligned}$$

- Q40. If $f(x) = \begin{cases} \sin(\cos x) + \cos(\sin x) & , x \in [-\frac{\pi}{2}, 0] \\ \sin(\lambda \cos x) - x + 1 & , x \in (0, \frac{\pi}{2}] \end{cases}$ has a maximum at $x = 0$, then number of all possible integral values of $\lambda \in [0, 2\pi]$ is :

Ans. 6

Sol.

$$\begin{aligned}
f(0) &= \sin 1 + -1 \\
f(0^+) &= \sin \lambda + 1 \\
f(0) > f(0^+) &\Rightarrow \sin \lambda < \sin 1 \\
\Rightarrow \lambda &= 0, 1, 3, 4, 5, 6
\end{aligned}$$

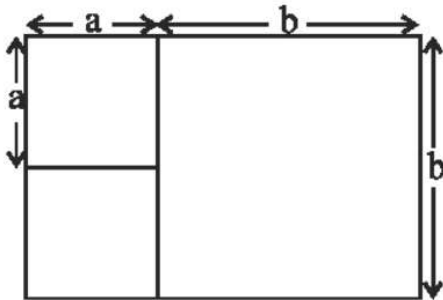
- Q41. For any 2 squares, the sum of whose area is 1, a rectangle of area A exists into which the squares can be placed without overlapping of interior points (Assume that the squares are to be placed with their sides parallel to the sides of the rectangle). The least value of A can be expressed as $\frac{1 + \sqrt{\alpha}}{\beta}$ where α and β are natural numbers, then $(\alpha + \beta)$.

Ans. 4

Sol. ($b > a$)

$$a^2 + b^2 = 1 \quad (1)$$

$$A = (a + b)b = a\sqrt{1 - a^2} + (1 - a^2)$$



$$\frac{dA}{da} = 0 \Rightarrow a^2 = \frac{2 - \sqrt{2}}{4} \text{ and } b^2 = \frac{2 + \sqrt{2}}{4} \Rightarrow A = \frac{1 + \sqrt{2}}{2}$$

- Q42. The sum of max. and the min. values of $f(x) = x^3 - 3x$ subject to condition $x^4 + 36 \leq 13x^2$.

Ans. 0

Sol. Given condition is

$$x^4 + 36 - 13x^2 \leq 0$$

$$x^4 - 13x^2 + 36 \leq 0$$

$$(x^2 - 9)(x^2 - 4) \leq 0$$

⚠ Possibly misplaced — please review:

$$(x + 3)(x - 3)(x + 2)(x - 2) \leq 0$$

$$x \in [-3, -2] \cup [2, 3]$$

$$\text{Now, } f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3$$

For max or min $f'(x) = 0$ gives

$$x^2 - 1 = 0$$

$$x = \pm 1$$

$$\text{Now, } f(1) = 1 - 3 = -2$$

$$f(-1) = -1 + 3 = 2$$

$$f(2) = 8 - 6 = 2$$

$$f(-2) = -8 + 6 = -2$$

$$f(3) = 27 - 9 = 18$$

$$f(-3) = -27 + 9 = -18$$

Thus, the maximum value of $f(x)$ is 18 at $x = 3$ and the minimum value of $f(x)$ is -18 at $x = -3$.

Q43. Let $f(x) = -4e^{2\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ and g be inverse function of f and $h(x) = \frac{a + bx^2}{x^{\frac{5}{3}}}$,

$h'(5) = 0$, then

Ans. 5

Sol. $g(f(x)) = x$

$$g'(f(x)) f'(x) = 1$$

$$f(x) = -7/6$$

$$\therefore x = 1$$

$$g'(-7/6)f'(1) = 1$$

$$g'\left(-\frac{7}{6}\right) = \frac{1}{f'(1)}$$

$$f'(x) = -4 \cdot e^{\frac{1-x}{2}} \left(-\frac{1}{2}\right) + x^2 + x + 1$$

$$f'(1) = 2 + 1 + 1 + 1 = 5$$

$$h(x) = ax^{-\frac{5}{3}} + bx^{\frac{1}{3}}$$

$$h'(x) = -\frac{5a}{4}x^{-\frac{9}{4}} + \frac{b}{4}x^{-\frac{3}{4}}$$

$$h'(5) = 0$$

$$\Rightarrow -\frac{5a}{4} \cdot 5^{-\frac{9}{4}} + \frac{b}{4} \cdot 5^{-\frac{3}{4}}$$

$$= 0$$

$$\Rightarrow 5a \cdot 5^{-\frac{3}{2}} = b$$

$$\Rightarrow \frac{a}{b} = 5^{\frac{1}{2}}$$

$$\left(\frac{a}{b}\right)^2 = 5$$

⚠ Possibly misplaced — please review:

$$\frac{a^2}{5b^2g'(-\frac{7}{6})} = \frac{5}{5 \times \frac{1}{5}} = 5$$

Q44. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and let $g(x) = f^{-1}(x)$. Find $g'''(0)$.

Ans. 1

Sol. Sol.

$$f(0) = 0, f'(0) = 1, f''(0) = 1, f'''(0) = 2$$

$$g(f(x)) = x$$

$$\Rightarrow g'(f(x))f'(x) = 1$$

$$\Rightarrow g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$$

$$\Rightarrow g'''(f(x))f'(x)$$

$$= - \left[\frac{(f'(x))^3 f'''(x) - 3(f''(x))^2 (f'(x))^2}{(f'(x))^6} \right]$$

Put $x = 0$

$$g'''(0)1 = - \left[\frac{1 \times 2 - 3 \times 1}{1} \right] = 1$$

- Q45. Let $y = f(x)$ be a differentiable function such that $f(1) = 1, f(2) = -3, f(3) = 4, f(4) = -1$ and $f(5) = 2$. Let $y = g(x)$ be a differentiable function such that $f'(x)g(x) \neq g'(x)f(x)$ then minimum number of real roots of the equation $(g'(x))^2 + g(x)g''(x) = 0$ is _____.

Ans. 4

Sol. $f(x) = 0$ has at least 4 real roots

$f(x) \cdot g(x) \neq g'(x)f(x) \Rightarrow g(x) = 0$ has at least 3 real roots

$\Rightarrow g'(x) \cdot g(x) = 0$ has at least 5 real roots

$\Rightarrow (g'(x))^2 + g(x)g''(x) = 0$ has at least 4 real roots

- Q50. Consider a circle with unit radius whose chord AB is at a perpendicular distance x from centre C. Let area of triangle ACB is given by function $y = f(x)$. On the curve $y = f(x)$ a variable point $P(x, f(x))$ is taken and perpendiculars PQ and PR are drawn on x & y axis respectively. Area of rectangle QPRO is given by $y = g(x)$. (where 'O' is origin).

Column-I		Column-II	
(a)	The greatest area of ACB is	(p)	$\frac{2}{3\sqrt{3}}$
(b)	The greatest area of rectangle QPRO is	(q)	$\frac{\sqrt{13}-1}{6}$
(c)	The greatest vertical distance between $y = f(x)$ & $y = g(x)$ occurs at x equal to	(r)	$\frac{1}{3}$
(d)	The area bounded by $y = f(x), y = 0$ & $x = 1$ is	(s)	$\frac{1}{2}$

Choose the correct option :

(A) $a \rightarrow p; b \rightarrow s; c \rightarrow q; d \rightarrow r$ (B) $a \rightarrow p; b \rightarrow q; c \rightarrow s; d \rightarrow r$

(C) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow s$ (D) $a \rightarrow s; b \rightarrow p; c \rightarrow q; d \rightarrow r$

Ans. D

Sol. a. $f(x) = x\sqrt{1-x^2}, 0 \leq x \leq 1$

$$f'(x) = \frac{(1 - \sqrt{2}x)(1 + \sqrt{2}x)}{\sqrt{1-x^2}}$$

$$f_{\max}(x) = \frac{1}{2}. \text{ Range } \left[0, \frac{1}{2}\right]$$

b. $g(x) = xf(x)$

$$g(x) = x^2\sqrt{1-x^2}$$

$$g'(x) = \frac{x(2-3x^2)}{\sqrt{1-x^2}}$$

$$\text{Range} \left[0, \frac{2}{3\sqrt{3}} \right]$$

$$c. \quad \because f(x) > g(x)$$

$\therefore$ Vertical distance

$$h(x) = f(x) - g(x) = \sqrt{1-x^2} (x-x^2)$$

$$h'(x) = \frac{(1-x)(1-x-3x^2)}{\sqrt{1-x^2}}$$

$$\Rightarrow \text{Greatest value occurs at } x = \frac{\sqrt{13}-1}{6}$$

$$d. \quad \text{Required Area} = \int_0^1 x\sqrt{1-x^2} dx = \frac{1}{3}$$

Q51. Let f be a polynomial of degree 4 having real coefficients satisfying

$$f'(0) = f'(1) = f'(-1) = 0 \text{ and } f(0) = 4, f''\left(\frac{1}{2}\right) = -1$$

Match the following column :

Column-I	Column-II
(a) $f(x) = 0$ has	(p) root at $x = 2$
(b) $4 - f(x) = 0$ has	(q) root at $x = 1$
(c) $f'(x) + x - 1 = 0$ has	(r) 2 equal real roots
(d) $xf'(x) - 4f(x) = 0$ has	(s) no real roots

Choose the **CORRECT** option :

$$(A) \text{ a-s; b-r; c-q,r; d-p} \quad (B) \text{ a-p; b-q; c-p; d-s} \quad (C) \text{ a-p,r,s; b-q,s; c-p; d-r}$$

$$(D) \text{ a-q; b-p; c-s; d-r}$$

Ans. A

Sol. Using the conditions, we get $f(x) = x^4 - 2x^2 + 4$

Q46. A differentiable function $f : R \rightarrow R$ satisfies $f(1) = 2, f(2) = 3$ and $f(3) = 1$. Show that $f'(x) = 0$ for some x .

Ans. -

Sol. Since f is differentiable, it is continuous. By the intermediate value theorem, there is a $c \in (2, 3)$ with $f(c) = 2 = f(1)$. Now by Rolle's theorem there is $x \in (1, c)$ with $f'(x) = 0$.

OR

The continuous function f over the closed interval $[1, 2]$ must attain its absolute maximum, which cannot be at either endpoint (since $f(2) > f(1)$ and $f(2) > f(3)$). So the maximum must be at an interior point x and then $f'(x) = 0$.

OR

By the mean value theorem, $f'(y) = 1 > 0$ for some $y \in (1, 2)$ and $f'(z) = -2 < 0$ for some $z \in (2, 3)$. So $f'(x) = 0$ for some $x \in (y, z)$ since for a differentiable f , the function f' satisfies the

intermediate value property by Darboux's theorem. (This is important to say because f' need not be continuous.)

- Q47. Let $f(x) = \frac{x^4}{(x-1)(x-2)\dots(x-n)}$ where the denominator is a product of n factors, n being a positive integer. It is also given that the X-axis is a horizontal asymptote for the graph of f . Answer the independent questions below by choosing the correct option from the given ones.
- How many vertical asymptotes does the graph of f have?
(A) n (B) less than n (C) more than n (D) impossible to decide
 - What can you deduce about the value of n ?
(A) $n < 4$ (B) $n = 4$ (C) $n > 4$ (D) impossible to decide
 - As one travels along the graph of f from left to right, at which of the following points is the sign of $f(x)$ guaranteed to change from positive to negative?
(A) $x = 0$ (B) $x = 1$ (C) $x = n - 1$ (D) $x = n$
 - How many inflection points does the graph of f have in the region $x < 0$?

Ans. (a) A (b) C (c) C (d) more than 1

Sol. (a) n , at $x = 1, 2, \dots, n$.
 (b) $n > 4$, because $\lim_{x \rightarrow \pm\infty} f(x) = 0$ and for this to happen, the degree of the denominator of $f(x)$ must be greater than that of the numerator.
 (c) $x = n - 1$, because $f(x)$ is positive for $x > n$ and $f(x)$ changes sign precisely when it passes through $x = 1, 2, \dots, n$. Note that the sign of $f(x)$ for $x < 0$ and for $x \in (0, 1)$ depends on the parity of n .
 (d) more than 1. Note that $f(x) = 0$ only at $x = 0$, with multiplicity 4. Without loss of generality, let n be even. (If n is odd, the reasoning is completely parallel, see note at the end.) Now $f(x) > 0$ for $x < 1$ except at $x = 0$ and f has all derivatives for $x < 1$. Due to the multiple root at $x = 0$, the graph of f must be concave up (i.e. $f''(x) > 0$) near $x = 0$. Further, as $x \rightarrow -\infty$, the values of $f(x)$ stay positive and $\rightarrow 0$. Therefore, as one traces the graph leftward from the origin, it must become concave down at least once and eventually concave up again so as to approach the X-axis from above. (Note: If n is odd, $f(x) < 0$ for $x < 1$ except at $x = 0$. As one traces the graph leftward from the origin, the function is initially as well as eventually concave down and must be concave up at least once in-between so as to approach the X-axis from below.)
 xcept at $x = 0$. As one traces the graph leftward from the origin, the function is initially as well as eventually concave down and must be concave up at least once in-between so as to approach the X-axis from below.)

- Q48. Consider the function $f(x) = ax + \frac{1}{x+1}$, where a is a positive constant. Let L = the largest value of $f(x)$ and S = the smallest value of $f(x)$ for $x \in [0, 1]$. Show that $L - S > \frac{1}{12}$ for any $a > 0$.

Ans. -

Sol. Let $f(x) = ax + \frac{1}{x+1}$. We wish to understand the minimum and maximum of this function in the interval $[0, 1]$.
 1]. Now $f(0) = 1$, $f(1) = a + \frac{1}{2}$ and $f'(x) = a - \frac{1}{(x+1)^2}$. Over the interval $[0, 1]$ the value of $f'(x)$ increases from $a - 1$ at $x = 0$ to $a - \frac{1}{4}$ at $x = 1$.

We should consider what happens to the sign of $f'(x)$. For this we consider the following cases.

(1) Suppose $a \leq 1/4$. Because $1/(x+1)^2 \geq 1/4$ on the interval $[0, 1]$, $f'(x) \leq 0$, so the maximum is at 0 and the minimum is at $x = 1$. So the difference is $1 - (1/2 + a) = 1/2 - a \geq 1/4 \geq 1/12$

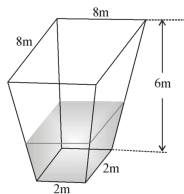
(2) Suppose $a \geq 1$. Then $f'(x) \geq 0$ on the interval $[0, 1]$, so maximum is at 1 and minimum at 0. We get $a + 1/2 - 1 = a - 1/2 \geq 1/2 \geq 1/12$.

(3) Suppose $1/4 \leq a \leq 1$. Now $f'(x) = 0$ at $\tilde{x} = \frac{1}{\sqrt{a}} - 1$. For this range of a , $\tilde{x} \in [0, 1]$. In the interval $[0, \tilde{x}]$, $f'(x) \leq 0$ and in the interval $[\tilde{x}, 1]$, $f'(x) \geq 0$. Now we make two sub-cases depending on at which endpoint the maximum occurs.

(3i) Suppose $1/4 \leq a \leq 1/2$. Then $f(0) \geq f(1)$. So minimum is at $\tilde{x}$, maximum is at $x = 0$. $f(\tilde{x}) = \sqrt{a} - a + \sqrt{a} = 2\sqrt{a} - a$. So the difference between maximum and minimum is $1 + a - 2\sqrt{a} = (1 - \sqrt{a})^2$. This is smallest when $\sqrt{a}$ is closest to 1 and so $xa + yb$, where the integers x, y can be found explicitly by backward substitution in the steps used to calculate the GCD. If a and b are coprime, i.e. if their GCD is 1, then we can write $1 = xa + yb$. This tells you that xa is 1 modulo b and yb is 1 modulo a . Therefore, $sxa + ryb$ is r modulo a and s modulo b .

The relevance for this problem is that the same reasoning applies for polynomials in one variable, because in this setting too one has division with remainder. Because x^{100} and $(x - 2)^3$ do not share a common factor, you know without any work that a polynomial with given properties must exist. The same algorithm as the previous paragraph (but now with polynomials) gives a systematic way to find it. In the given problem we could use a different trick because the specified remainders here were rather simple (constants). But there is a conceptual way as well by implementing the Chinese remainder theorem.

- Q49. A pond has been dug at the IIT Bombay as an inverted truncated pyramid with a square base (see figure below). The depth of the pond is 6 m. The square at the bottom has side length 2 m and the top square has side length 8 m. Water is filled in at a rate of $\frac{19}{3}$ cubic meters per hour. At what rate is the water level rising exactly 1 hour after the water started to fill the pond?



Ans. -

Sol. Denote by $V(t)$ the volume of the water after t hours the water started to fill the pond. Then $V(t) = \frac{19}{3}t$ (cubic meters). Let $h(t)$ be the level of the water and $a(t)$ be the side of the square at the top of the water after t hours the water started to fill the pond.

Extend the truncated pyramid to get the non-truncated pyramid and let x be the height of the added part.

Then we have the following picture

Similarity of triangles implies relations

$$\frac{x}{x + h(t)} = \frac{2}{a(t)}$$

$$\frac{x}{x + 6} = \frac{2}{8}$$

Multiplying them we find

$$\frac{x}{x+6} = \frac{1}{4}, x = 2.$$

Hence,

$$a(t) = h(t) + 2$$

Now,

$$\begin{aligned} V(t) &= \frac{h(t)}{3}(4 + a(t)^2 + 2a(t)) = \frac{h(t)}{3}(4 + h(t)^2 + 4h(t) + 4 + 2h(t) + 4) = \\ &= \frac{h(t)}{3}(h(t)^2 + 6h(t) + 12) = \frac{1}{3}((h(t) + 2)^3 - 8) \end{aligned}$$

Differentiate this :

$$V'(t) = (h(t) + 2)^2 h'(t)$$

Then

$$h'(1) = \frac{V'(1)}{(h(1) + 2)^2} = \frac{V'(1)}{(3V(1) + 8)^3} = \frac{19}{3} \cdot \frac{1}{9} = \frac{19}{27} \text{ (meters per second)}$$

